Theory / ASIC Layout
(Perfect Symmetry)

Source 2

Source 1

Length = X
Capacitance = C

Length = X
Capacitance = C

Dest 2

Dest 1

Balanced Power Trace
(Hiding Succeeds)

FPGA Reality
(Automated Switch Matrix Routing)

Source 1
(LUT)

Rigid FPGA Routing Fabric

Source 2
(LUT)

Switch
Matrix 1

Congested Route
Length = X+Δ
Cap = C2 (C1 ≠ C2)

Switch
Matrix 2

Direct Route
Length = X
Cap = C1

Switch
Matrix 3

Dest 1
(LUT)

Congested Route
Length = X+Δ
Cap = C2 (C1 ≠ C2)

Dest 2
(LUT)

Transient Glitch Leakage
(|t| = 56.98, Hiding Fails)